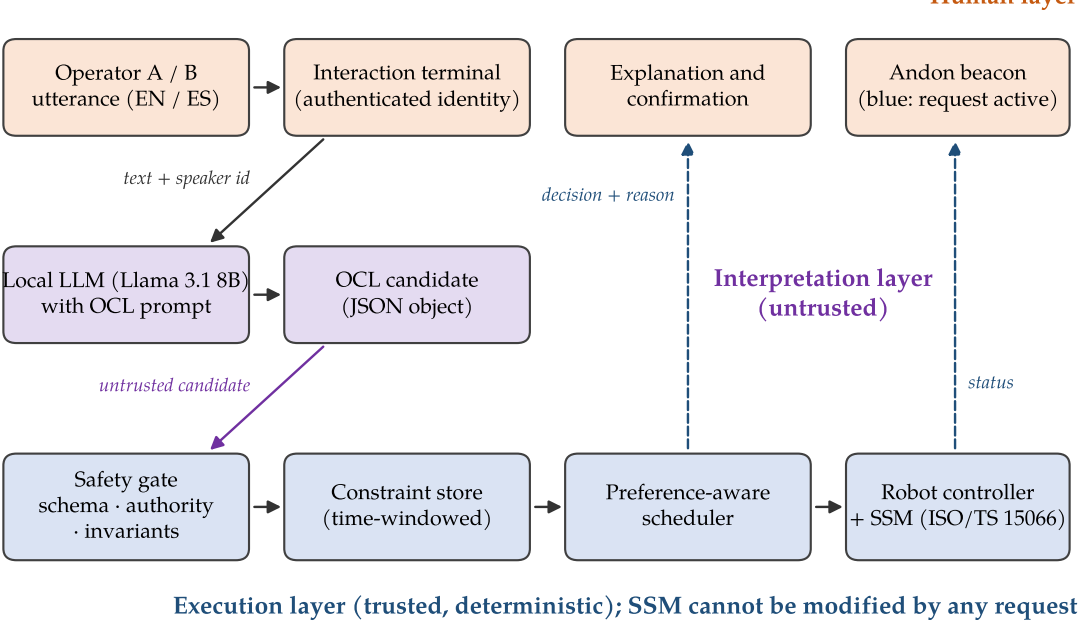
Human layer
Operator A / B utterance (EN / ES)
Interaction terminal (authenticated identity)
Explanation and confirmation
Andon beacon (blue: request active)
text + speaker id
decision + reason
Local LLM (Llama 3.1 8B) with OCL prompt
OCL candidate (JSON object)
Interpretation layer (untrusted)
untrusted candidate
status
Safety gate schema · authority · invariants
Constraint store (time-windowed)
Preference-aware scheduler
Robot controller + SSM (ISO/TS 15066)
Execution layer (trusted, deterministic); SSM cannot be modified by any request